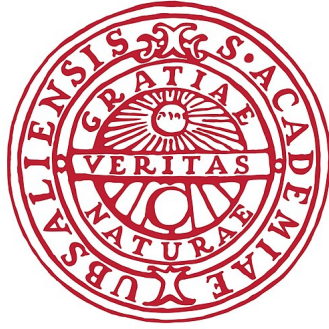




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Uncertainty-Aware Survival Modeling of Pavement Deterioration with Connected Vehicle Data

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ABSTRACT

Survival models are commonly used to predict when pavements will need maintenance, yet the uncertainty surrounding these predictions is rarely quantified. This gap poses challenges for road agencies that must make multi-million-dollar decisions based on point estimates without knowing how reliable these predictions are.

This study propagates parameter uncertainty into time-to-maintenance predictions using high-frequency connected vehicle (CV) data. Data were collected weekly from 3,707 Swedish road sections (3,357 deterioration events) and analyzed using Cox and Weibull models with identical covariate structures. CV derived indicators were treated as time-dependent covariates, and prediction uncertainty was quantified using section-level bootstrap resampling.

Including CV indicators improved model fit and provided additional information beyond static road attributes. The Weibull model produced a median time-to-maintenance of 17.7 weeks with a 95% bootstrap confidence interval of 16.9–18.7 weeks. In contrast, the Cox model could not estimate a median within the study period, illustrating that the predicted maintenance timing strongly depends on model specification.

Overall, the results show that uncertainty quantification is essential for risk-aware pavement management. The framework developed here supports a shift from deterministic scheduling to probabilistic, uncertainty-aware planning, enabling more informed and efficient maintenance decisions.

KEYWORDS

Bootstrap

Cox proportional hazards models

International Roughness Index

Parametric models

Pavement Management System

Semi-parametric models

Weibull distribution model

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1 INTRODUCTION

1.1 The Pavement Management Problem

A pavement management system (PMS) is a structured decision framework used by road authorities to plan, prioritize, and optimize maintenance and rehabilitation activities. A PMS typically requires the agency to (i) monitor pavement condition, (ii) predict future deterioration, (iii) schedule interventions, and (iv) allocate long-term budgets efficiently (Haas et al., 2015). These systems apply to all types of pavement, including asphalt, concrete, gravel, and composite structures.

Pavement deterioration is driven by interacting factors such as traffic loading, climate, material properties, and past maintenance histories (Haas et al., 2015). Because these processes evolve stochastically over time, the moment when a pavement section reaches an unacceptable condition level is inherently uncertain. This uncertainty has direct economic importance: intervening too late increases rehabilitation costs and safety risks, while intervening prematurely leads to inefficient use of scarce maintenance budgets. Understanding and predicting this timing is therefore central to effective infrastructure management.

Traditional Modeling Approaches

Conventional pavement condition assessment relies on periodic surveys performed using profiling vehicles, laser measurement systems, or visual inspections. These surveys are typically conducted at multi-year intervals (Federal Highway Administration, 2016), producing discrete snapshots of pavement performance that can be related to traffic volumes and environmental covariates.

Duration models (survival analysis), estimate the time until an event occurs and provide a natural framework for analyzing such time-to-maintenance problems. Survival models explicitly accommodate right-censored road sections (those that have not yet deteriorated by the end of the study period) and can relate the deterioration timing to observable characteristics. In Swedish pavement studies, survival models have been applied to evaluate the lifetime of asphalt binders and to predict the timing of maintenance actions at the network level (Eklöf and Wendel, 2018; Eklöf, 2020). These applications demonstrate that duration models can capture both heterogeneity across road segments and uncertainty in deterioration timing, making them

suitable tools for maintenance planning.

The Opportunity and Challenge of Connected Vehicle Data

Recent advances in telematics have enabled the use of connected vehicles (CV) as mobile pavement-sensing platforms. Modern vehicles are equipped with accelerometers, GPS units, and other embedded sensors that generate high-frequency data during routine driving (Gao et al., 2019). Pavement roughness indicators, especially the International Roughness Index (IRI) can be derived from these data streams and have been shown to correlate with traditional survey measurements (Bektas et al., 2020). These CV derived indicators offer substantially higher temporal and spatial resolution than periodic condition surveys.

However, CV data also introduce new complexities. Measurements vary by vehicle type, sensor placement, and driving behavior (Cheng et al., 2018). Sampling is inherently uneven, reflecting traffic patterns rather than controlled survey design (Paterson and Scullion, 2019). These features mean that CV derived indicators differ fundamentally from static covariates: they are noisy, irregularly sampled, and time-dependent. Incorporating them into duration models requires methods capable of handling both measurement variability and dynamic covariates.

Uncertainty in Survival Predictions

Although survival models have been widely applied in pavement engineering, most studies emphasize point estimates such as mean survival times or hazard ratios. In practice, infrastructure decisions depend not only on the estimated deterioration timing but also on the uncertainty surrounding these predictions. Sources of uncertainty include variability in CV derived covariates and sampling differences across road sections. However, applied pavement studies rarely quantify how these uncertainties propagate into predicted maintenance times, despite the fact that maintenance planning involves substantial financial commitments (Pascual and Meeker, 1999; Collett, 2015). This gap motivates the development of modeling approaches that provide both predictions and explicit measures of their reliability.

1.2 Aim and Research Questions

The aim of this thesis is to examine how model formulation and data characteristics influence the stability and reliability of time-to-maintenance predictions. In particular, the analysis explores three related questions: how estimated hazard rates and survival functions change when connected vehicle (CV) indicators are included compared with models using only static road covariates; how predictive uncertainty differs between semi-parametric and parametric survival formulations; and how sensitive the predicted maintenance times are to uncertainty in model parameters. Together, these questions guide the evaluation of both model behaviour and decision-relevant uncertainty in pavement deterioration forecasting.

1.3 Contribution of This Study

This thesis contributes to pavement deterioration modeling in several ways. First, it integrates high-frequency connected vehicle (CV) indicators into duration models as time-dependent covariates, allowing for continuous tracking of roughness signals over weekly intervals. Second, it provides a comparative analysis of Cox proportional hazards and Weibull specifications under identical covariate structures, enabling a clear assessment of how model choice influences both predicted hazard patterns and time-to-maintenance estimates. Third, it explicitly propagates parameter uncertainty through the predictive pipeline, moving beyond traditional reliance on point estimates and demonstrating how uncertainty affects maintenance timing. Finally, the thesis employs section-level bootstrap resampling to capture sampling variability while preserving within-section dependence across recurrent deterioration episodes.

1.4 Thesis Outline

The thesis proceeds as follows. Chapter 2 reviews the relevant literature on pavement management, survival analysis, and connected vehicle data. Chapter 3 describes the data and the methodological framework. Chapter 4 presents empirical results structured around the three research questions. Chapter 5 discusses the findings, implications, limitations, and conclusions.

2 LITERATURE REVIEW

This chapter situates the thesis within three bodies of work: (i) pavement management and deterioration modeling, (ii) duration (survival) analysis as a tool for maintenance timing, and (iii) the use of connected vehicle (CV) data for condition monitoring. Then it synthesizes what is known about integrating CV indicators into duration models and identifies the specific gaps addressed in this study.

2.1 Pavement Management and Deterioration Modeling

Pavement management systems (PMS) support network-level planning by linking condition monitoring to intervention timing and budgeting (Haas et al., 2015). Deterioration dynamics reflects traffic loads, climate, materials and maintenance history, implying substantial variability between sections and over time (Haas et al., 2015). To inform timing decisions in a resource-constrained setting, modeling approaches have evolved from deterministic rules to probabilistic frameworks. Traditional deterministic methods include the AASHTO empirical design equations, which predict a single pavement service life based on traffic, materials, and structural design assumptions, and the Mechanistic–Empirical Pavement Design Guide (MEPDG), which uses mechanistic stress–strain calculations combined with empirical calibration to forecast pavement performance. Although more detailed than the AASHTO equations, MEPDG still produces point predictions of time-to-maintenance and does not quantify uncertainty. In contrast, probabilistic frameworks explicitly represent variability and heterogeneity in deterioration timing (ARA, Inc., 2004; World Bank, 2018; Micevski et al., 2002).

2.2 Duration Analysis for Maintenance Decisions

Duration (survival) analysis models the distribution of time T from an origin to an event; in this setting, the time until a pavement section requires maintenance, while correctly handling right-censoring for sections that have not deteriorated by the end of follow-up and accommodating time-varying explanatory variables. Fundamental quantities include the survival function $S(t) = P(T > t)$, the hazard $h(t)$, and the cumulative hazard $H(t)$, which are standard objects in the statistical literature (Kalbfleisch and Prentice, 2002; Klein and Moeschberger, 2003). In this thesis, the event of interest is the first passage of a section’s roughness across the inter-

vention threshold (e.g., $IRI \geq \tau$) during a spell, where a spell corresponds to one maintenance cycle from its beginning until intervention or censoring. The duration T is measured in weeks from the start of observation for each spell until the event, and if no event occurs during follow-up the spell is right-censored. Because sections may experience multiple deterioration cycles, durations recur across spells and the connected vehicle (CV) indicators evolve over time, motivating the use of time-dependent covariates. Parametric specifications such as Weibull impose a functional form for the baseline hazard, whereas semi-parametric Cox models estimate relative risks without specifying the baseline (Lawless, 2003; Cox, 1972). Duration analysis is particularly appropriate for pavement maintenance because decision variables concern *when* thresholds are crossed, many sections are unfailed at the end of the study, and condition indicators including CV derived signals change over time.

Maintenance datasets also exhibit features that require careful risk-set construction. Beyond right-censoring and time-varying covariates, they often include repeated spells for the same section across multiple intervention cycles and delayed entry (left truncation), because a section first appears in the data at the time of its initial sensing or survey rather than at the beginning of the true deterioration cycle. Addressing these features requires a start–stop (counting-process) data structure, which we implement in Chapter 3.

Nonparametric Kaplan–Meier (KM) curves provide a descriptive view of the empirical time-to-maintenance distribution without imposing functional assumptions and answer the question: “*What fraction of sections remain below the intervention threshold by time t ?*” KM estimates are useful for summarising the observed deterioration pattern and diagnosing whether the hazard increases or decreases with time, which informs the choice of parametric baseline shapes. However, KM curves do not adjust for covariates and cannot incorporate time-varying predictors, making them a descriptive starting point rather than a predictive model (Klein and Moeschberger, 2003). Because they reveal broad hazard patterns prior to modeling and serve as a benchmark for comparison, they are generally used first in empirical analysis.

Duration methods have been applied to pavement performance at both material and network levels: for example, to evaluate lifetimes of binders and mixes (Eklöf and Wendel, 2018) and to inform scheduling and policy analysis through models incorporating traffic and environmental covariates (Prozzi and Madanat, 2004; Yang et al., 2019). Duration models naturally handle censoring and accommodate time-varying indicators when the data are structured appropriately (Mishalani and Madanat, 2002). However, much of the existing applied work has focused on

point predictions such as hazard ratios or median lifetimes and has given limited attention to uncertainty in predicted time-to-maintenance, the quantity most relevant for decision-makers. Sections 2.4–2.5 return to this deficiency when establishing the research gap for the present study.

2.3 Connected Vehicle Data for Condition Monitoring

Traditional pavement condition surveys using profilers, laser measurement systems, or visual inspections provide accurate but infrequent measurements, typically at intervals of one to three years (Federal Highway Administration, 2016). In contrast, connected vehicle (CV) sensing uses everyday vehicles as mobile platforms, generating high-frequency data streams from accelerometers, GPS units, and other telematics during routine driving (Gao et al., 2019). From these data, roughness indicators such as the International Roughness Index (IRI) can be derived and have shown good agreement with survey-based measures at network scales (Bektas et al., 2020; Eboli et al., 2017). Because agencies often trigger interventions based on condition classes (e.g., IRI thresholds), CV derived indicators have direct operational relevance for identifying deterioration events. In this thesis, the intervention threshold itself defines the deterioration event, meaning that a section is considered to have deteriorated at the point where its IRI first exceeds the agency’s intervention criterion (Sayers and Karamihas, 1998; Federal Highway Administration, 2016).

CV signals differ from traditional static covariates in several important statistical respects. First, measurements are noisy and heterogeneous across vehicles, sensor configurations, and speeds, raising classical measurement-error concerns that can bias coefficient estimates and understate predictive uncertainty if not addressed (Carroll et al., 2006; Cheng et al., 2018). Second, the sampling of CV data is irregular in time and space because it is driven by traffic patterns rather than a fixed survey plan; as a result, information density varies across sections and weeks (Rizopoulos, 2012; Paterson and Scullion, 2019). Third, multiple high-frequency measurements must be aggregated to the analysis interval (weekly), introducing an additional layer of variability (Li et al., 2020). Together, these characteristics mean that CV indicators are inherently time-dependent covariates whose variability can materially influence survival estimation and prediction.

2.4 Integrating CV Indicators with Duration Models: What is Known and Missing

Studies that integrate richer, more frequent condition information into survival models show that time-dependent covariates can improve predictive performance relative to static specifications (Yang and Gong, 2021). However, the specific properties of CV data measurement variability, irregular sampling, and aggregation effects remain underexplored within the survival framework. The applied literature typically reports uncertainty for parameters (e.g., hazard ratios) but not for the *predicted time-to-maintenance* that drives planning, even though that is the quantity of practical interest (Collett, 2015; Meeker and Escobar, 1998). Consequently, it is unclear how much predictive uncertainty arises from (i) model choice (semi-parametric vs. parametric), (ii) sampling variability across sections and (iii) the variability of CV indicators themselves (Omer et al., 2021).

2.5 Research Gap and Positioning of This Thesis

Integrating the existing literature reveals several gaps that motivate this present study. Although connected vehicle (CV) data have gained attention in recent years, there remains limited empirical evidence on how high-frequency, noisy CV indicators influence hazard and survival estimates when incorporated as time-dependent covariates in pavement deterioration models (Hong and Prozzi, 2010; Yang and Gong, 2021). Much of the applied work continues to highlight parameter uncertainty and rarely extends uncertainty quantification to the decision-relevant quantity of interest, namely time-to-maintenance leaving practitioners without calibrated prediction intervals (Collett, 2015; Meeker and Escobar, 1998). A further unresolved issue concerns the relative contribution of different uncertainty sources: it is not clear whether predictive uncertainty arises primarily from model specification choices (such as Cox versus Weibull), from sampling variability at the section level, or from the inherent variability of CV indicators themselves (Omer et al., 2021; Carroll et al., 2006).

Positioned against these gaps, this thesis incorporates CV indicators as time-dependent covariates within Cox and Weibull formulations using a counting-process structure, allowing dynamic modeling of deterioration signals. It contrasts predictive behaviour across model classes under identical covariate sets to isolate the role of specification choice. Finally, it propagates uncertainty to decision-relevant outputs through section-level bootstrap resampling that pre-

serves within-section dependence. Together, these components define how the thesis addresses the limitations in the existing literature and contributes to a clearer understanding of predictive performance and uncertainty in pavement deterioration modeling.

3 DATA AND METHODOLOGY

This chapter describes the data and the modeling strategy used to analyse pavement deterioration as a time-to-event process. The dataset consists of weekly observations for each road section, allowing us to incorporate connected vehicle (CV) indicators as time-dependent covariates within survival analysis. We proceed from data description and event definition to a unified explanation of spells, delayed entry, and the counting-process formulation. We then present the Kaplan–Meier, Cox, and Weibull modeling components followed by the bootstrap procedure for uncertainty quantification.

3.1 Data

We analyse road sections from two Swedish municipalities, Nacka and Östersund, each observed over 144 consecutive weeks (≈ 2.8 years). The analytical unit is the road section, indexed by $k = 1, \dots, K$. Each section contributes a sequence of weekly observations. In total, the raw files contain 206,040 section–week observations for Nacka (2,089 sections) and 186,393 observations for Östersund (1,759 sections).

Each observation contains three CV derived indicators `iri_mean` (mean International Roughness Index measured in meters per kilometre (m/km)), `variance_mean` (within-week roughness variability), and `r_count_mean` (mean number of contributing vehicles). For modeling, these indicators are standardized (mean–SD scaling) but retain their original names, appearing as “(scaled)” in Chapter 4. These quantities are computed from aggregated vehicle acceleration, speed, and GPS readings (Douangphachanh and Oneyama, 2014; Bektas et al., 2020; Zhao et al., 2021). To reduce simultaneity, indicators from week t are used to predict deterioration in week $t+1$.

The static road attributes consist entirely of categorical variables: municipality (Nacka, Östersund), Functional Class (2–5 and an “Other” category), Road Type (e.g., Major link, Bussficka, Huvudgata, Lokalgata, Parkering, Uppsamlingsgata, plus an “Other” category), and Surface Type (ABT, Asphalt, Grus, and “Other”). Some of these levels appear infrequently in the raw data. Rare categories defined as those with fewer than 50 sections or with no observed deterioration events are merged into the “Other” category to ensure stable estimation. This step prevents quasi-separation, which occurs when a category perfectly (or nearly perfectly) predicts the outcome and leads to unstable or infinite coefficient estimates in both Cox and

Weibull models.

3.2 Event definition and time scale

In this thesis, a pavement section is assumed to require maintenance the moment its roughness exceeds a predefined intervention threshold. Under this assumption, a “deterioration event” and an “intervention” coincide: the time at which the International Roughness Index (IRI) first crosses the threshold is treated as the time at which maintenance would be triggered. Let $\tau = 2.5$ m/km denote this threshold, consistent with common operational practice in pavement management (Sayers and Karamihas, 1998; Federal Highway Administration, 2016).

Formally, the event time is defined as the *first-passage time* at which a section’s weekly IRI transitions from $< \tau$ to $\geq \tau$. This first-passage definition ensures that the model captures the moment a section first enters an unacceptable condition state. Time is measured in weeks from the beginning of each observation spell.

Because weekly IRI readings may fluctuate around the threshold due to measurement noise or short-term variability in connected-vehicle signals, we also considered a robustness rule that requires two consecutive weeks with $\text{IRI} \geq \tau$ before registering an event. This robustness check serves only to verify that the main findings are not driven by isolated threshold spikes. In practice, the two definitions produced nearly identical event timings for the dataset, and therefore the primary analysis relies on the first-passage rule.

3.3 Spells, delayed entry, and the counting-process structure

Road sections may deteriorate multiple times within the study window. We adopt a *gap-time formulation*, which means that the survival clock resets after each deterioration event or maintenance reset. A spell begins at the start of a deterioration cycle and ends at the next threshold crossing or censoring. This formulation is appropriate when interest lies in the time until the next failure rather than the cumulative behaviour over the entire observation period.

A complication arises because sections may enter the dataset after their deterioration cycle has already begun, a feature known as *delayed entry* or *left truncation*. Let C_0 denote the (unobserved) time of the previous intervention, C_s the first calendar week the section appears in the data, and C_1 the next intervention time. The full spell duration is $Z = C_1 - C_0$, of which only the remaining portion $R = C_1 - C_s$ is observed; the elapsed but unobserved portion is

$U = C_s - C_0$, so that $Z = U + R$.

Valid inference requires the standard assumption that, conditional on covariates X , the sampling time does not convey additional information about the full spell duration:

$$C_s \perp Z \mid X \iff U \perp R \mid X,$$

where the symbol $\perp$ denotes conditional independence. Intuitively, after conditioning on X , being observed “early” or “late” in a spell should not be associated with unusually long or short remaining time. This assumption would be violated, for example, if data collection (or CV coverage) occurred disproportionately when sections were near failure.

Delayed entry and time-varying covariates together motivate a *counting-process (start–stop)* formulation. Each spell is represented as a sequence of weekly intervals $(t_{\text{start}}, t_{\text{stop}}]$ with an event indicator marking the end of the interval. A section contributes to the risk set only after its entry time C_s . The covariate vector $X_{kr}(t)$ contains both static attributes and the CV indicators for week t (lagged one week). This representation yields correct risk sets for the Cox model and a natural piecewise-constant structure for parametric likelihoods.

After cleaning and merging sparse categories, the analytical dataset contains $K = 3,707$ sections and 3,357 deterioration events across spells. The underlying data exhibit substantial heterogeneity: some sections deteriorate frequently, while others remain below the threshold throughout the 144 week study spell.

3.4 Modeling strategy

The modeling process proceeds through three components. First, Kaplan–Meier (KM) survival curves provide a nonparametric description of the empirical time-to-deterioration distribution and reveal whether hazards appear to increase or decrease. KM serves as a descriptive reference, and conceptual details are given in Section 2.2.

Next, Cox proportional-hazards models relate covariates to the hazard without imposing a baseline hazard form (Cox, 1972). It is well-suited when interest is centered on relative risks and when testing whether CV indicators add predictive information beyond static attributes. A baseline model uses only static attributes.

Baseline (static attributes only).

$$h_k(t) = h_0(t) \exp\left(\beta_1 \text{Municipality}_k + \beta_2 \text{FunctionalClass}_k + \beta_3 \text{RoadType}_k + \dots + \beta_6 \text{Area}_k\right).$$

where $h_k(t)$ is the hazard of deterioration of road section k at time t and $h_0(t)$ is an unspecified baseline hazard function common to all sections.

The extended specification adds the time-dependent CV indicators, because the extended model nests the baseline (set $\theta=0$), so a likelihood-ratio test assesses whether CV adds information. Estimation uses partial likelihood with Breslow ties.

Let X_k^{static} denote the vector of time-invariant covariates for section k , $X_k^{\text{CV}}(t)$ the vector of time-dependent CV indicators at time t , and let β and θ be the corresponding parameter vectors; the superscript $\top$ denotes vector transpose.

Extended (nests the baseline by adding CV indicators).

$$h_k(t) = h_0(t) \exp(\beta^\top X_k^{\text{static}} + \theta^\top X_k^{\text{CV}}(t)),$$

where $X_k^{\text{CV}}(t)$ collects the time-dependent CV indicators (lagged one week). Temporal ordering is enforced by lagging the CV indicators so that $X_k^{\text{CV}}(t)$ predicts the hazard in the subsequent week. This avoids simultaneity between the measured condition and the event definition.

Finally, a Weibull model provides a fully parametric alternative with a monotone baseline hazard.

$$h_k(t) = \lambda_k \gamma t^{\gamma-1}, \quad S(t) = \exp(-\lambda t^\gamma), \quad \lambda_k(t) = \exp(\beta^\top X_k^{\text{static}} + \theta^\top X_k^{\text{CV}}(t)).$$

Covariates enter multiplicatively on the hazard scale, retaining a proportional-hazards interpretation. Time-dependent covariates are implemented piecewise over weekly intervals. For a given covariate profile, the median time-to-maintenance has the closed form

$$t_{0.5,k}(t) = (\log 2)^{1/\gamma} \exp(\beta^\top X_k^{\text{static}} + \theta^\top X_k^{\text{CV}}(t)),$$

and comparisons with KM curves assess the suitability of the Weibull form.

To compare parametric specifications, we use Akaike's Information Criterion (AIC), defined as

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p,$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood and p is the number of estimated parameters. AIC balances goodness of fit with model complexity: lower AIC values indicate a better trade-off between fit and parsimony. Because the Cox model is semi-parametric and does not provide a full likelihood, AIC comparison is applied only when comparing the baseline and extended Weibull models in Chapter 4.

3.5 Estimation and inference with time-dependent covariates

All survival models are estimated using a start–stop (counting-process) representation, in which each weekly interval contributes one observation with covariates held constant within the interval.

Cox model parameters are estimated by maximizing the partial likelihood. At an event time t_i , the contribution to the likelihood is

$$\frac{\exp(\beta^\top X_k^{\text{static}} + \theta^\top X_k^{\text{CV}}(t_i))}{\sum_{j \in \mathcal{R}(t_i)} \exp(\beta^\top X_j^{\text{static}} + \theta^\top X_j^{\text{CV}}(t_i))},$$

where $\mathcal{R}(t_i)$ is the risk set immediately before t_i . Standard errors are obtained from the inverse observed information matrix. The baseline hazard $h_0(t)$ is estimated post-hoc using the Breslow estimator (Breslow, 1972; Klein and Moeschberger, 2003).

For a given covariate profile X , the Cox model expresses the survival function as

$$S(t | X) = S_0(t)^{\exp(\beta^\top X)},$$

where $S_0(t)$ is the baseline survival function. The estimated baseline $\hat{S}_0(t)$ is obtained from the fitted partial likelihood using the Breslow estimator and takes the form of a step function that decreases only at observed event times (Andersen et al., 1993; Kalbfleisch and Prentice, 2002). This construction means that the Cox survival curve remains constant between event times and does not extrapolate beyond the observed follow-up period.

The Weibull model is estimated by maximum likelihood. With time-dependent covariates, the scale parameter $\lambda_k(t)$ is treated as a piecewise constant within each weekly interval. The likelihood contribution for an observed time t with the event indicator δ is

$$L_k(t, \delta) = [h_k(t)]^\delta S_k(t) = (\lambda_k \gamma t^{\gamma-1})^\delta \exp(-\lambda_k t^\gamma),$$

and survival contributions accumulate across weekly intervals for each section. The parameter vectors (β, θ) and the shape parameter γ are estimated jointly by maximizing the log-likelihood, with standard errors given by the inverse Hessian.

3.6 Uncertainty quantification (section-level bootstrap)

To quantify how parameter uncertainty propagates to predicted maintenance times, we implement a section-level bootstrap with $B = 300$ replications. In each replication, road sec-

tions are resampled with replacement, preserving all spells associated with each selected section. The extended Weibull model is re-estimated on each bootstrap sample to obtain draws $(\hat{\beta}^{(b)}, \hat{\theta}^{(b)}, \hat{\gamma}^{(b)})$, which are used to compute predicted medians for a reference covariate profile,

$$\hat{T}^{(b)} = (\log 2)^{1/\hat{\gamma}^{(b)}} \exp(\hat{\beta}^{(b)\top} X_{\text{ref}}^{\text{static}} + \hat{\theta}^{(b)\top} X_{\text{ref}}^{\text{CV}}).$$

The empirical distribution of $\{\hat{T}^{(b)}\}$ yields a bootstrap standard deviation and percentile-based confidence intervals. Pilot checks with $B = 150$ and $B = 300$ produced nearly identical endpoints, supporting the use of $B = 300$.

4 RESULTS

This chapter presents results in the following order: KM descriptives, Cox baseline and Cox + CV models, Weibull models for calendar-time prediction, and bootstrap-based uncertainty, with additional diagnostics in Appendix A.

4.1 Descriptive analysis

Figure 1 shows Kaplan-Meier survival by municipality. We can see that the KM survival function for Östersund is lower than that for Nacka across all follow-up periods. Both curves decline over time, reflecting the overall deterioration pattern in the dataset.

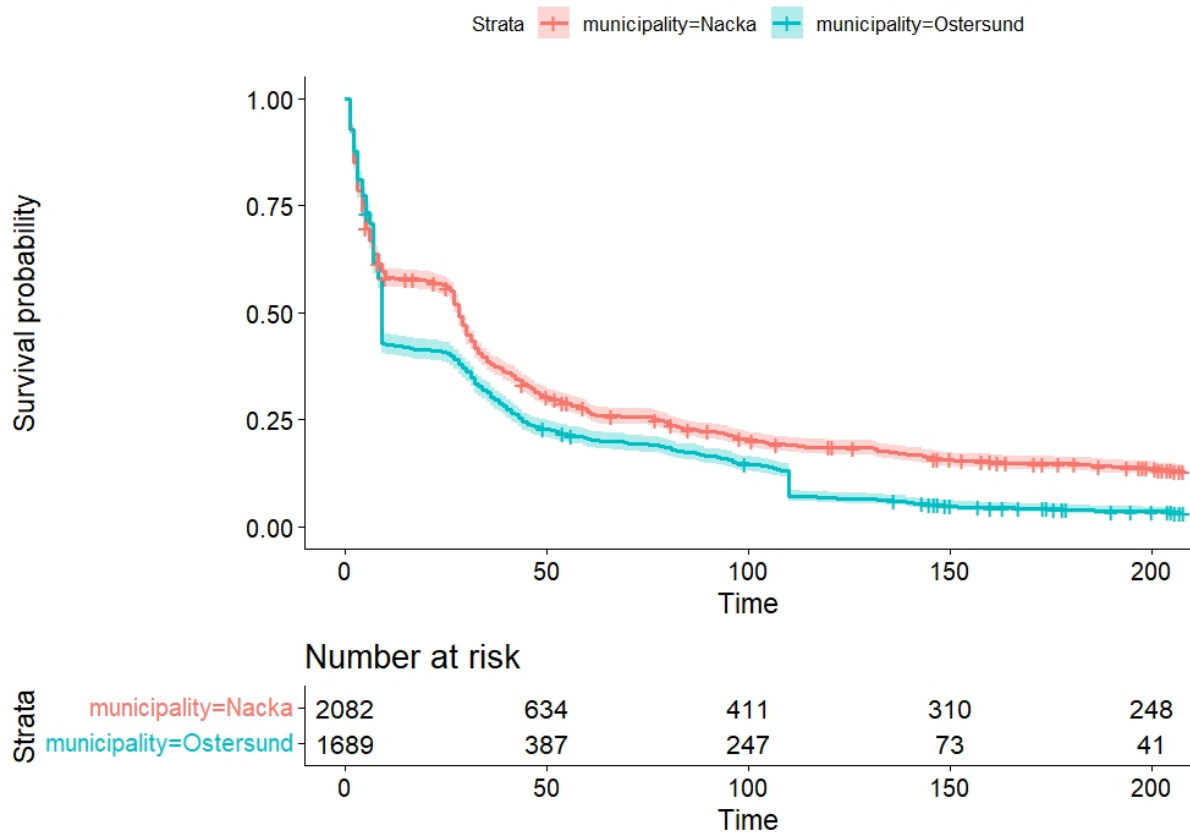


Figure 1: Kaplan-Meier survival by municipality (shaded: 95% Confidence Intervals).

4.2 Impact of CV indicators on hazard and survival

Table 1 reports hazard ratios (HR) for the baseline Cox model.

Table 1: Cox baseline model with static covariates. Reference levels: Municipality = Nacka; Functional Class = 2; Road type = Major link.

Variable	HR	95% CI	<i>p</i> – value
Municipality (Östersund vs. Nacka)	6.34	(2.64 - 15.26)	< 0.001
Functional Class 3	1.88	(1.37 - 2.58)	< 0.001
Functional Class 4	1.75	(1.29 - 2.36)	0.000293
Functional Class 5	2.77	(2.06 - 3.72)	< 0.001
Functional Class Other	0.86	(0.31 - 2.39)	0.768
Road type: Bussficka	2.47	(1.52 - 4.00)	0.000262
Road type: Huvudgata	1.79	(1.38 - 2.31)	< 0.001
Road type: Kantstensparkering	5.89	(3.95 - 8.79)	< 0.001
Road type: Lokalgata	4.83	(3.40 - 6.88)	< 0.001
Road type: Lokalgata liten	1.19	(0.91 - 1.55)	0.194
Road type: Other	2.72	(1.82 - 4.08)	< 0.001
Road type: Parkering	5.15	(3.45 - 7.68)	< 0.001
Road type: Uppsamlingsgata	3.37	(2.34 - 4.85)	< 0.001
Surface type: ABT	1.07	(0.91 - 1.25)	0.409
Surface type: Asfalt	1.60	(0.67 - 3.79)	0.287
Surface type: Grus	0.40	(0.16 - 1.00)	0.051
Surface type: Other	1.07	(0.79 - 1.44)	0.664
Length (scaled)	1.13	(1.03 - 1.24)	0.007
Area (scaled)	0.94	(0.85 - 1.03)	0.165

In line with the Kaplan–Meier survival curves in Figure 1, Table 1 shows that road sections in Östersund have a higher deterioration hazard than those in Nacka. In the context of this study, a higher hazard does not indicate worse road quality; rather, it means that sections reach the intervention threshold sooner and are therefore renovated more quickly.

Higher Functional Classes (3–5) are also associated with higher hazards, with Functional Class 5 showing the strongest effect. This pattern reflects prioritisation: higher-class roads are

typically monitored more closely and maintained earlier than lower-class roads.

Hazard ratios (HRs) quantify relative timing. For example, an HR greater than one indicates that a section is expected to reach the intervention threshold sooner than the reference category, while an HR below one indicates a longer expected time until renovation, holding other factors constant.

A positive association in this context means that a characteristic is linked to a shorter time until the intervention threshold is reached, that is, earlier renovation rather than poorer road quality. Consistent with this interpretation, section length shows a positive association with the hazard ($HR = 1.13$ per standard deviation), indicating that longer road sections tend to reach the intervention threshold sooner than shorter ones. Put differently, shorter road sections typically remain below the intervention threshold for a longer period before maintenance is triggered.

Table 2 shows the extended Cox model where we add CV time-dependent indicators; the likelihood-ratio test (LRT) compares extended vs baseline.

Table 2: Cox extended (adds CV time-dependent indicators).

Variable	HR	95% CI	<i>p</i> – value
Municipality (Östersund vs. Nacka)	4.55	(1.88 - 11.02)	0.000782
Functional Class 3	1.65	(1.20 - 2.28)	0.002
Functional Class 4	1.46	(1.07 - 1.99)	0.017
Functional Class 5	1.95	(1.42 - 2.67)	< 0.001
Functional Class Other	0.83	(0.30 - 2.31)	0.717
Road type: Bussficka	2.15	(1.32 - 3.50)	0.002
Road type: Huvudgata	1.73	(1.33 - 2.24)	< 0.001
Road type: Kantstensparkering	4.63	(3.07 - 6.97)	< 0.001
Road type: Lokalgata	3.73	(2.60 - 5.37)	< 0.001
Road type: Lokalgata liten	1.17	(0.90 - 1.54)	0.241
Road type: Other	2.34	(1.55 - 3.54)	< 0.001
Road type: Parkering	3.95	(2.62 - 5.95)	< 0.001
Road type: Uppsamlingsgata	2.63	(1.81 - 3.83)	< 0.001
Surface type: ABT	1.10	(0.94 - 1.29)	0.222
Surface type: Asfalt	1.53	(0.64 - 3.63)	0.337
Surface type: Grus	0.38	(0.15 - 0.96)	0.041
Surface type: Other	1.13	(0.83 - 1.52)	0.437
Length (scaled)	1.10	(1.00 - 1.21)	0.048
Area (scaled)	0.97	(0.88 - 1.07)	0.587
Variance_mean (scaled)	1.01	(0.98 - 1.05)	0.491
R_count_mean (scaled)	0.73	(0.64 - 0.83)	< 0.001

A likelihood-ratio test comparing the baseline and extended Cox specifications shows that including CV indicators improves model fit ($\chi^2(2) = 34.48$, $p < 0.001$), indicating that time-dependent CV information adds explanatory value beyond static road attributes.

Within the extended model, `r_count_mean` (scaled) exhibits a protective association (HR = 0.73, $p < 0.001$), indicating that sections with higher connected vehicle coverage tend to reach the intervention threshold later than comparable sections with lower coverage. In this

context, a positive association (i.e., a hazard ratio greater than one) would imply earlier renovation, whereas a protective association (hazard ratio less than one) implies delayed renovation. Accordingly, higher traffic or CV coverage is associated with lower prioritisation for near-term rehabilitation rather than better intrinsic road quality, reflecting monitoring intensity or maintenance prioritisation rather than a causal effect. The `variance_mean` (scaled) does not show a statistically meaningful association.

Including CV indicators also reduces the municipality effect (HR reduced from 6.34 to 4.55), suggesting that part of the observed spatial heterogeneity is captured by differences in traffic exposure or condition information.

Rather than focusing on formal proportional-hazards diagnostics, we emphasise the implications for prediction. While the Cox model is useful for assessing relative differences in deterioration timing, it does not yield a finite in-sample median time-to-maintenance for the reference profile. This motivates comparison with a fully parametric Weibull specification, which allows explicit calendar-time predictions such as median maintenance intervals.

4.3 Cox vs Weibull

Table 3 summarises the extended Weibull model, and Table 4 compares model fit using Akaike's Information Criterion (AIC).

Table 3: Weibull (extended): coefficients and shape parameter γ .

Parameter	Estimate	SE	$p - value$
Intercept	7.226	0.705	< 0.001
Municipality (Östersund)	-2.378	0.651	0.00026
Functional Class 3	-0.848	0.235	0.00031
Functional Class 4	-0.627	0.228	0.0059
Functional Class 5	-1.007	0.232	< 0.001
Functional Class Other	0.309	0.755	0.683
Road type: Bussficka	-1.258	0.359	0.00046
Road type: Huvudgata	-0.930	0.190	< 0.001
Road type: Kantstensparkering	-2.480	0.300	< 0.001
Road type: Lokalgata	-2.126	0.267	< 0.001
Road type: Lokalgata liten	-0.289	0.198	0.144
Road type: Other	-1.319	0.302	< 0.001
Road type: Parkering	-2.212	0.301	< 0.001
Road type: Uppsamlingsgata	-1.569	0.276	< 0.001
Surface type: ABT	-0.136	0.115	0.236
Surface type: Asfalt	-0.588	0.638	0.357
Surface type: Grus	1.512	0.676	0.025
Surface type: Other	-0.111	0.222	0.618
Length (scaled)	-0.137	0.069	0.045
Area (scaled)	0.017	0.071	0.810
Variance_mean (scaled)	-0.009	0.028	0.741
R_count_mean (scaled)	0.516	0.096	< 0.001
Shape γ	0.694	0.009	—

Table 4: Weibull AIC comparison.

Model	AIC
Baseline Weibull	30827.48
Extended Weibull	30790.84

A lower AIC indicates a better balance between goodness of fit and model complexity. The extended Weibull specification achieves a substantially lower AIC than the baseline Weibull model, indicating that including CV indicators improves model fit. The estimated Weibull shape parameter is $\hat{\gamma} = 0.694 < 1$, which implies that the deterioration hazard decreases over time. This pattern is consistent with the empirical survival curves shown by the Kaplan–Meier estimates. Importantly, the advantage of the Weibull model in this application is not related to proportional-hazards validity but to its ability to produce explicit calendar-time predictions.

While the Cox model is well suited for estimating relative differences in deterioration timing, it does not yield a finite median time-to-maintenance in this dataset because the estimated survival curve for the reference profile remains above 0.5 throughout the observation period. In contrast, the Weibull model specifies the baseline hazard parametrically, which allows direct calculation of calendar-time quantities such as the median time-to-maintenance.

Under the extended Weibull specification, the predicted median time-to-maintenance for the reference profile is 17.71 weeks. This median is estimable because the Weibull model extrapolates the survival curve beyond the observed follow-up, whereas the Cox model does not in Figure 2. For this reason, the Weibull specification is preferable when the objective is to obtain explicit time-to-maintenance predictions rather than relative risk comparisons.

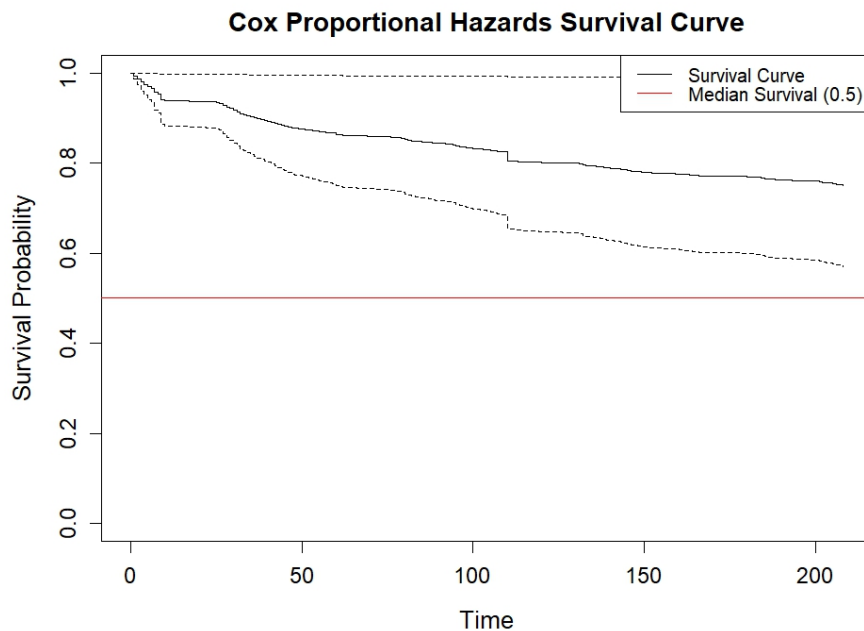


Figure 2: Cox survival for the reference profile; curve remains above 0.5 over the study horizon, so the median is non-estimable.

Cox captures empirical patterns but yields no in-study median here (survival > 0.5); Weibull provides a finite median with bootstrap uncertainty: 17.71 weeks (95% CI [16.87, 18.65]).

4.4 Sensitivity to parameter uncertainty (bootstrap)

With section-level bootstrap $B = 300$, the distribution of predicted medians has mean = 17.71, SD = 0.482, 95% CI = [16.87, 18.65] (Figure 3). The Q-Q plot used to assess the normality of the bootstrap distribution is provided in Appendix A.2, Figure 6.

Table 5: Bootstrap summary for predicted median time-to-maintenance (weeks).

Statistic	Weeks
Mean	17.71
Median	17.71
SD	0.482
95% CI	[16.87 - 18.65]

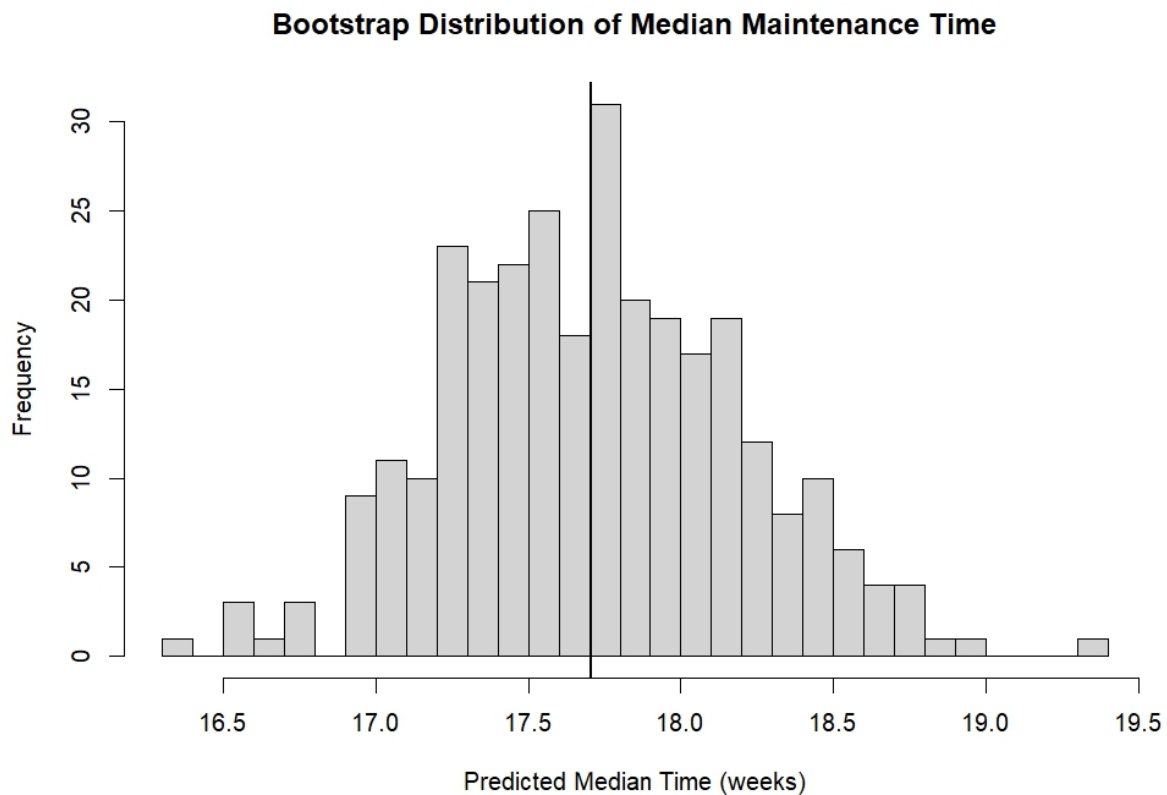


Figure 3: Bootstrap distribution of predicted median time-to-maintenance ($B = 300$).

Relative uncertainty is modest:

$$\frac{SD}{mean} = \frac{0.48}{17.71} \approx 2.7\%;$$

we also report the 95% lower bound (≈ 16.87 weeks) for conservative planning. The bootstrap distribution is approximately normal with low dispersion, indicating stable parameter estimates.

5 DISCUSSION AND CONCLUSIONS

This chapter interprets the empirical findings (Chapter 4) in light of the aims and research questions, outlines methodological and practical implications, and concludes with limitations, avenues for future work, and overarching conclusions.

5.1 Interpretation of Main Findings

The findings show that integrating connected vehicle (CV) derived indicators, high-frequency measures of pavement roughness and vehicle coverage, meaningfully enhances predictive performance relative to static only Cox specifications (LRT $\chi^2(2) = 34.48, p < 0.001$). The protective association for `r_count_mean (scaled)` (HR = 0.73, $p < 0.001$) should not be interpreted causally; rather, it likely reflects prioritisation of high-traffic links or design features correlated with traffic intensity. Meanwhile, the non-significant effect of `variance_mean (scaled)` suggests that the average roughness level captured by the CV indicators is more informative for predicting deterioration timing than short-term week-to-week fluctuations, which do not show a statistically meaningful association in the model. These results indicate that high-frequency CV coverage provides decision-relevant information beyond static covariates while requiring careful interpretation (see Chapter 4, Tables 1–2).

Model specification also plays a fundamental role in shaping predictions. The Cox model captures relative differences in deterioration timing based on the observed data, but it does not specify the baseline hazard parametrically. As a result, when the estimated survival curve for the reference profile remains above 0.5 over the observed follow-up period, the median time-to-maintenance cannot be determined from the Cox model. By contrast, the Weibull model imposes a monotone baseline hazard and yields a closed-form median of 17.71 weeks (95% CI [16.87–18.65]). This contrast highlights a core trade-off: the Cox model offers empirical flexibility, whereas the Weibull model provides predictive completeness useful for planning under uncertainty (see Chapter 4, Sections 4.3 and 4.4).

Finally, the parameter uncertainty associated with these predictions is modest. Section-level bootstrapping ($B = 300$) produces a narrow distribution of medians (SD ≈ 0.482 weeks; relative uncertainty $\approx 2.7\%$), with a distribution that is approximately symmetric and bell-shaped, indicating that the bootstrap estimates are well behaved (Appendix A.2, Figure 6). These intervals, however, are conditional on the model and do not reflect additional uncertainty sources

such as measurement error or spatial dependence that would broaden prediction bands (see Chapter 4, Section 4.4). Together, these findings demonstrate that CV indicators enrich predictive modeling, model choice materially affects decision quantities, and parameter uncertainty, although modest remains an important component of uncertainty-aware maintenance planning.

5.2 Methodological implications

The methodological differences between the Cox and Weibull models help explain why they produce fundamentally different types of prediction. Because Cox baseline survival is a step function, the estimated survival curve for a given profile can remain flat over long periods of follow-up. When this happens, as in the reference profile shown in Figure 2, the survival curve may stay above 0.5 throughout the observation window, making the median time-to-maintenance non-estimable without imposing additional assumptions about the survival tail. This is not a flaw of the Cox model but a direct consequence of its design: the model avoids specifying the baseline hazard shape and is therefore best suited for estimating relative effects rather than producing calendar-time predictions.

In contrast, the Weibull model assumes a monotone baseline hazard and provides closed-form predictions for quantities such as medians. The estimated shape parameter $\hat{\gamma} = 0.694 < 1$ indicates a decreasing hazard over time, consistent with the Kaplan–Meier curves. By imposing a parametric form, the Weibull model delivers decision-ready quantities such as median times and confidence intervals, though this comes at the cost of stronger assumptions that must be evaluated against nonparametric diagnostics.

5.3 Practical implications for pavement management

The results from this study have several practical implications for pavement management. First, connected vehicle indicators provide valuable information beyond traditional static inventories, helping agencies identify sections where deterioration risk is elevated even after accounting for structural and contextual factors. In practice, `r_count_mean (scaled)` serves as an informative proxy for coverage or traffic exposure rather than a causal determinant of deterioration, and can therefore assist agencies in prioritising sections for monitoring and intervention. Second, uncertainty-aware planning is essential: Weibull-based median predictions, supplemented with bootstrap confidence intervals (e.g., 17.71 weeks; 95% CI [16.87 – 18.65]),

support risk-informed scheduling rather than reliance on single-point forecasts. In operational terms, maintenance planners can use lower confidence bounds to construct conservative intervention schedules or examine interval widths to identify locations where further monitoring would yield the greatest reduction in decision uncertainty. Finally, model choice should be aligned with specific operational goals. Cox models are well suited for understanding relative risks and identifying influential factors, whereas parametric models such as Weibull provide explicit calendar-time predictions needed for budgeting and planning horizons. In practice, the two classes of models can be combined: Cox for covariate screening and Weibull (or other parametric forms) for generating actionable predictions.

5.4 Limitations

Several limitations should be acknowledged when interpreting the findings of this study. The analysis is based on data from two Swedish municipalities, and the generalisability of the results to other regions, climates, or maintenance times remains uncertain. Additionally, CV indicators are subject to measurement variability arising from differences in vehicles, speeds, and sensor configurations; because measurement error was not modeled explicitly, the reported prediction intervals likely understate total uncertainty. Another limitation concerns within-section dependence: the analysis treats recurring deterioration episodes from the same road section as conditionally independent once covariates are accounted for, which may understate correlation between episodes from the same section. Residual non-proportionality also persists in the Cox models, affecting the interpretation of time-constant hazard ratios for some covariates.

5.5 Future research

Future research could extend this work in several practical directions. First, incorporating additional sources of condition data or applying the model to road networks in other municipalities would help assess the generalisability of the results. Second, explicitly accounting for measurement uncertainty in the CV indicators, such as variation across vehicles or sensing conditions, may provide a more complete picture of predictive uncertainty. Third, evaluating alternative parametric survival models, such as the log-normal or gamma families, would help assess the sensitivity of the predictions to assumptions about the baseline hazard. Finally, comparing deterioration patterns across different climates, road types, or maintenance regimes would offer

further insight into how context influences model performance and maintenance timing.

5.6 Conclusions

This thesis develops an uncertainty-aware survival modeling framework that integrates connected vehicle (CV) indicators as time-dependent covariates and provides decision-ready predictions of time-to-maintenance. Three main conclusions follow. First, CV indicators improve predictive performance relative to static-only models, with `r_count_mean` (scaled) showing a protective association, meaning that sections with higher connected vehicle coverage tend to reach the intervention threshold later than comparable sections with lower coverage. Second, model specification has direct implications for decision-making: the Cox model offers robust relative-risk assessment but cannot provide an in-study median when $S(t) > 0.5$, whereas the Weibull model yields finite, interpretable medians (17.71 weeks; 95% CI [16.87–18.65]) suitable for scheduling and budgeting. Third, uncertainty can and should be made explicit. Section-level bootstrap resampling produces compact, interpretable intervals, allowing agencies to plan more effectively under uncertainty.

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A APPENDIX: ADDITIONAL DIAGNOSTICS

A.1 Schoenfeld Residual Plots

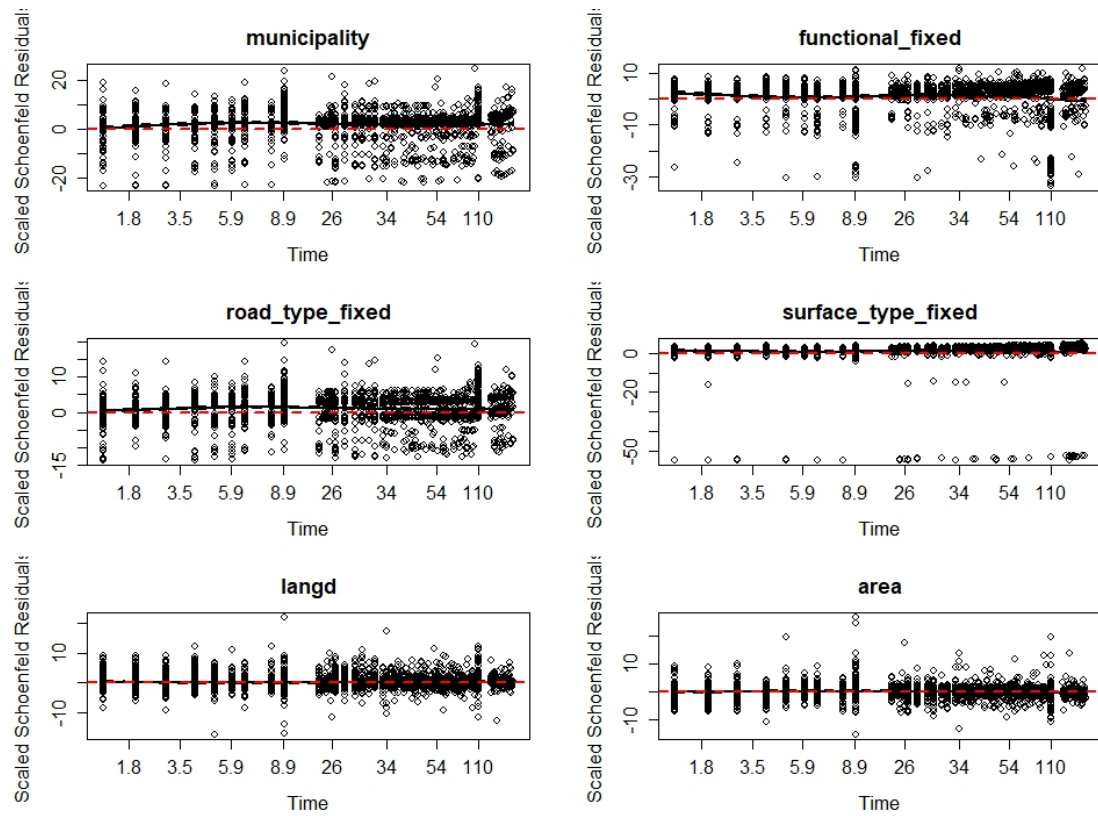


Figure 4: Scaled Schoenfeld residuals—baseline Cox model

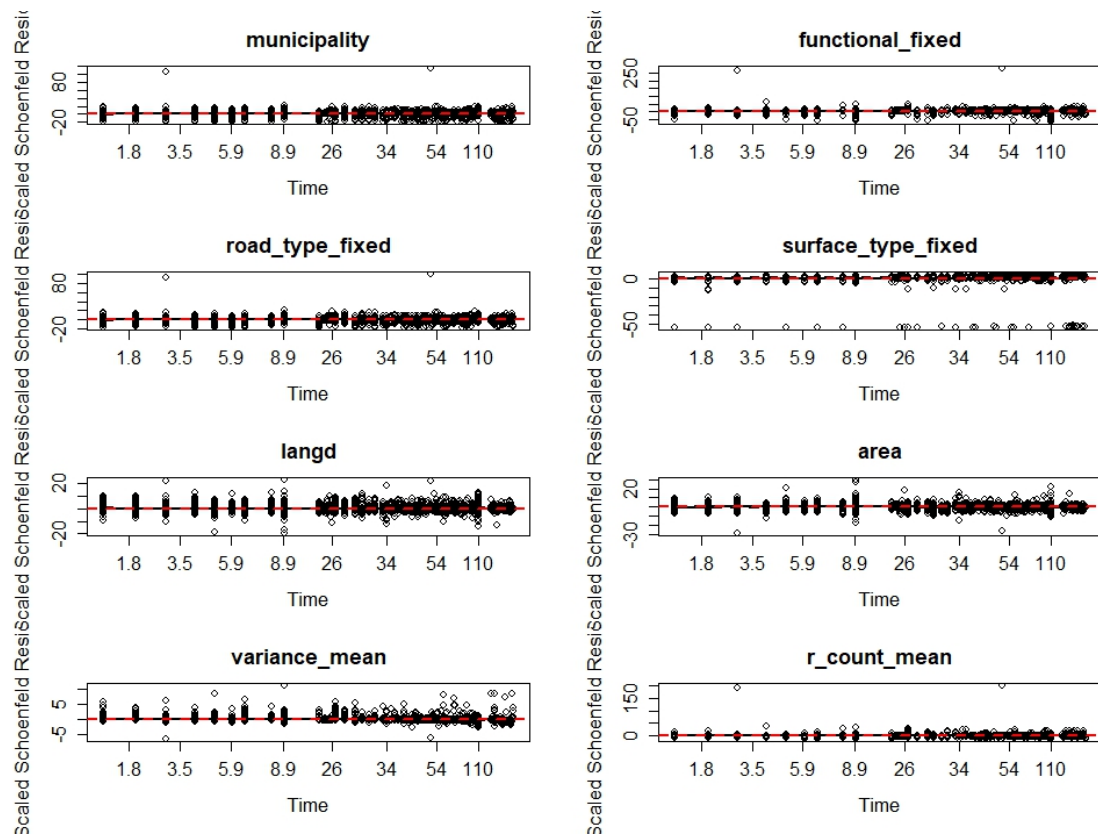


Figure 5: Scaled Schoenfeld residuals—extended Cox model

A.2 Bootstrap Diagnostics

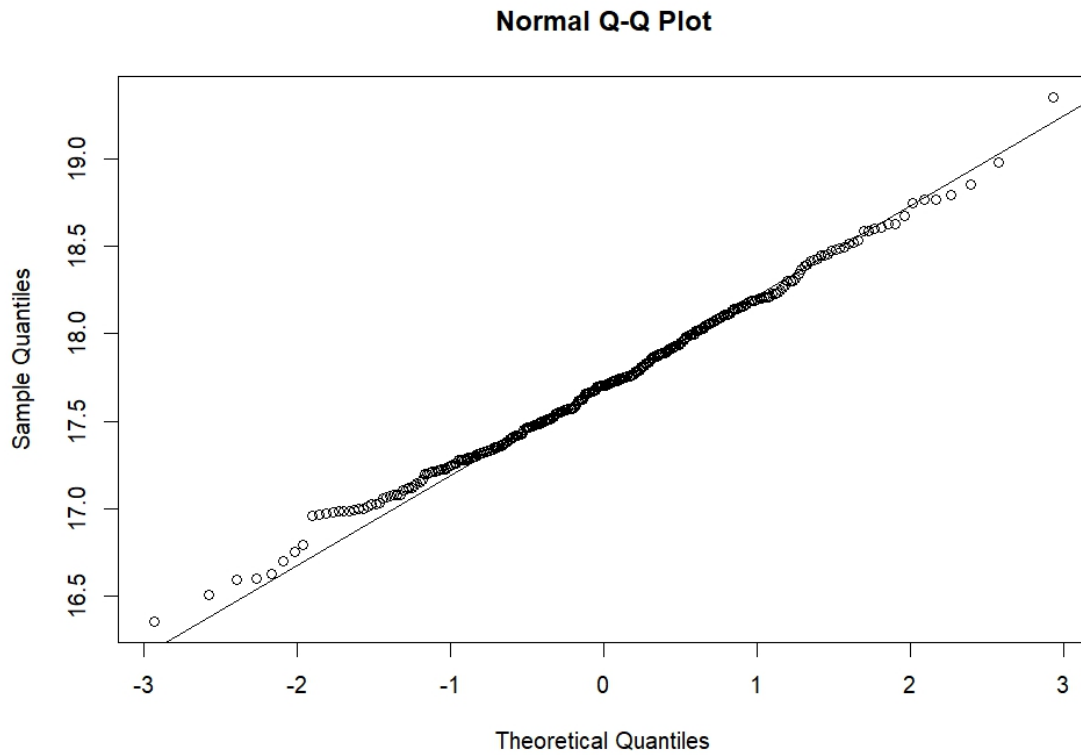


Figure 6: Q–Q plot of bootstrap medians

A.3 Full Proportional Hazards Test Results

Table 6: Complete PH test results—baseline Cox

Variable	χ^2	df	<i>p</i> – value
Municipality	18.176	1	< 0.001
Functional class	25.324	4	< 0.001
Road type	34.011	8	< 0.001
Surface type	23.593	4	< 0.001
Length	0.113	1	0.740
Area	0.180	1	0.670
Global	51.555	19	< 0.001

Table 7: Complete PH test results—extended Cox

Variable	χ^2	df	<i>p</i> – value
Municipality	17.247	1	< 0.001
Functional class	32.883	4	< 0.001
Road type	35.361	8	< 0.001
Surface type	22.556	4	< 0.001
Length	0.005	1	0.942
Area	0.002	1	0.968
Variance_mean (scaled)	4.661	1	0.031
R_count_mean (scaled)	4.576	1	0.032
Global	57.508	21	< 0.001